

Phase 9 CVRP Suite Report

Overview

- Condition count: 6.
- Best held-out condition: phase9_closeout_baseline_clarke_wright_savings.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Accepted Epochs	Fallback Epochs	Generation Errors	Mean Novelty	Mean Complexity
phase9_closeout_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	73.12826	0	0	0	0.0	0.0
phase9_closeout_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	77.9822	0	0	0	0.0	0.0
phase9_closeout_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1306.26666	0	0	0	0.0	0.0
phase9_closeout_direct_generate_plus_one_repair	direct_generate_plus_one_repair	0.0	3.0	n/a	225.9288	1	0	0	0.975	13.99
phase9_closeout_budget_matched_no_replay	budget_matched_no_replay	1.0	7.513251	7.513251	330.62868	1	0	0	0.973976	13.8
phase9_closeout_replay_solver_evolution	replay_solver_evolution	1.0	0.273435	0.273435	142.37906	0	0	0	0.0	0.0

Condition Notes

phase9_closeout_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 73.12826.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.

- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_closeout_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.064783.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 77.9822.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_closeout_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1306.26666.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.086899.
- Worst held-out instances:
- X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]

- X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_closeout_direct_generate_plus_one_repair

- Execution mode: direct_generate_plus_one_repair.
- Train feasibility rate: 0.0.
- Train penalized gap: 3.0.
- Held-out feasibility rate: 0.0.
- Held-out penalized gap: 3.0.
- Held-out runtime ms: 225.9288.
- Accepted epoch count: 1.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.0.
- Worst held-out instances:
 - X-n176-k26: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
 - X-n190-k8: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
 - X-n223-k34: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
 - X-n247-k50: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
 - X-n275-k28: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']

phase9_closeout_budget_matched_no_replay

- Execution mode: budget_matched_no_replay.
- Train feasibility rate: 1.0.
- Train penalized gap: 6.033943.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 7.513251.
- Held-out runtime ms: 330.62868.
- Accepted epoch count: 1.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 2.149073.
- Worst held-out instances:
 - X-n190-k8: feasible=True, penalized gap=18.523086, errors=[]
 - X-n275-k28: feasible=True, penalized gap=6.395152, errors=[]
 - X-n223-k34: feasible=True, penalized gap=4.616341, errors=[]
 - X-n176-k26: feasible=True, penalized gap=4.558103, errors=[]

- X-n247-k50: feasible=True, penalized gap=3.473574, errors=[]

phase9_closeout_replay_solver_evolution

- Execution mode: replay_solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 142.37906.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
 - X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
 - X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
 - X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

Judge Note

Phase-9 CVRP Closeout Suite Summary (Conservative)

Condition	Feasibility	Heldout Feasible Gap	Runtime (ms)	Robustness Dispersion	Notes
Baseline Clarke-Wright Savings	100%	**0.0738**	78	0.0078	Best heldout condition overall
Baseline Nearest Neighbor Constructive	100%	0.2734	73	0.0028	Faster, but larger gap than Clarke-Wright
Baseline Regret Insertion Local Search	100%	0.4664	1306	0.087	Much slower, worse gap, less robust
Direct Generate + One Repair	0%	-	226	0	Zero feasibility, penalized gap=3.0 (failed approach)
Budget-Matched No Replay	100%	7.5133	331	2.1491	Feasible but very large gap and low robustness
Replay Solver Evolution	100%	0.2734	142	0.0028	Better than budget-matched no replay; gap equals N.N.

Key Findings

- **Direct synthesis (Direct Generate + One Repair)** failed completely on feasibility and objective.

- **Budget-matched independent search without replay** had good feasibility but showed very poor objective gaps (worst gap of 7.5) and high robustness dispersion, indicating low reliability.
- **Replay-aware iterative search (Replay Solver Evolution)** maintained full feasibility, improved robustness, and significantly better objective gap (0.2734) compared to no-replay control, but did **not** beat Clarke-Wright baseline.
- The **fixed-baseline Clarke-Wright Savings method** clearly outperformed all learned methods in objective gap (0.0738) and matched 100% feasibility.
- The **Nearest Neighbor baseline** was faster and robust but with a larger gap.
- The **Regret Insertion Local Search baseline** was slow, less robust, and had worse objective quality.

Conclusion

- No learned condition clearly improved upon the best fixed baseline (Clarke-Wright Savings).
- Replay-aware iterative search outperformed the budget-matched no-replay learned approach in gap, robustness, and runtime.
- Fixed baselines remain strong baselines for phase-9 CVRP closeout.